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LABELS AND ANNOTATIONS

Files

SG AND 6G LAB

LAB EXPERIMENT

New Folder

ASSEMENT QNO 1.m

EX : 2.m

ex 1.m

EX 2 .m

EX 2.m

Ex 2.m

ex 5.m

ex1.m

Workspace

Name	Value	Size	Class
Capacity	4×16 double	4×16	double
Gain	128	1×1	double
i	4	1×1	double
j	16	1×1	double
Nt	[8,32,64,128]	1×4	double
SNR	1×16 double	1×16	double
SNRdB	1×16 double	1×16	double

untitled4.m

EX4.m

EX6.m

EX7.m

EX8.m

EX9.m

```
1 clc; clear; close all;
2
3 % Massive MIMO Capacity vs SNR and Gain
4
5 Nt = [8 32 64 128]; % Number of Base Station Antennas
6 SNRdB = 0:2:30; % SNR Range (dB)
7 SNR = 10.^(SNRdB/10);
8
9 Capacity = zeros(length(Nt), length(SNR));
10
11 for i = 1:length(Nt)
12
13     Gain = Nt(i); % Massive MIMO Array Gain
14
15     for j = 1:length(SNR)
16
17         % Shannon Capacity with Array Gain
18         Capacity(i,j) = log2(1 + Gain*SNR(j));
19
20     end
21
22 end
```

Figure 1

Massive MIMO Capacity vs SNR and Array Gain

SNR (dB)	8 Antennas	32 Antennas	64 Antennas	128 Antennas
0	3.5	5.0	6.0	7.0
10	7.5	10.0	11.5	13.0
20	11.5	14.0	16.0	18.0
30	15.5	18.0	20.0	22.0

Command Window

>>

32°C Mostly cloudy

Search

ENG IN

11:42 05-08-2026